Darshan Patel

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Summary

Computational Data Science MS student (Temple University, GPA 3.72) with production ML and analytics experience across 110M+ records. Built NLP models reaching 87.36% accuracy, hybrid RAG systems (Recall@5 0.856), and forecasting pipelines across 30K+ time series. Skilled in Python, SQL, machine learning, A/B testing, and turning data into business recommendations.

Education

Temple University **Expected May 2027**
Master of Science in Computational Data Science (GPA: 3.72 / 4.0) *Philadelphia, PA*
- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Knowledge Discovery and Data Mining, Deep Learning

BVM Engineering College **2021 – 2025**
Bachelor of Technology in Computer Engineering *Anand, India*
- **Relevant Coursework:** Data Structures and Algorithms, Database Management Systems, Probability and Statistics, Neural Networks

Technical Skills

Languages: Python, SQL, R, SAS

Data Analysis & Reporting: Data Cleaning & Validation, Statistical Inference, A/B Testing & Power Analysis, Cohort & Funnel Analysis, Dashboards (Tableau, Power BI, Streamlit), Microsoft Excel & PowerPoint

Machine Learning: Predictive Modeling, Time-Series Forecasting, NLP, LLMs & RAG, Scikit-Learn, LightGBM, XGBoost, PyTorch, Hugging Face

Data Tools: DuckDB, MySQL, MongoDB, Pandas, NumPy, SciPy, Statsmodels, Parquet, Git, Docker, AWS S3

Experience

IndiaNIC Infotech **Jan 2025 – May 2025**
Machine Learning Intern *Ahmedabad, India*
- Fine-tuned a hybrid RoBERTa-BiLSTM architecture with hierarchical attention across 200K samples, achieving 87.36% 4-class accuracy (+11% over LSTM baselines).
- Served real-time inference via a containerized Flask REST API integrating confidence thresholding for out-of-distribution inputs, structured logging, and emotion-triggered automated actions.
- Built an operational analytics dashboard tracking SKU-level emotion and sentiment trends, helping teams spot products drawing negative customer reactions.

Meritorious Infotech **May 2024 – Jun 2024**
Software Engineering Intern *Surat, India*
- Developed 10+ reusable, data-driven ReactJS components to streamline client-side state handling and reduce frontend feature development cycles by 20%.
- Integrated RESTful APIs to ingest, validate, and parse dynamic JSON payloads, optimizing asynchronous rendering performance and cross-platform load times.

Projects

E-commerce Growth Analytics | *SQL, DuckDB, Python, Streamlit* **2026**
- Built a tested SQL pipeline that cleaned and validated 110M records with 10 automated data-quality checks (duplicates, invalid values, revenue reconciliation), retaining 99.65% of data.
- Traced an 18.6% weekly conversion drop through a metric tree and segment decomposition; a daily drill-down revealed a likely checkout or tracking outage (1 purchase from 762K sessions).
- Delivered an 8-view dashboard and a one-page memo with three quantified recommendations; modeled customer lifetime value for 433K customers (3% holdout error vs. 52% for a baseline).

MedGuide: Local Hybrid RAG Health Q&A | *Ollama, ChromaDB, Sentence-Transformers, Streamlit* **2026**
- Architected a fully offline, zero-cost clinical Q&A system over 10K+ NIH MedQuAD passages using Ollama (Llama 3.2 3B) and ChromaDB.
- Developed a hybrid retrieval pipeline combining dense vector embeddings with BM25 lexical search via Reciprocal Rank Fusion, achieving a Recall@5 of 0.856.
- Designed grounded-refusal guardrails to mitigate medical hallucinations, evaluated via automated LLM-as-a-judge benchmarking across out-of-domain queries.

Walmart Sales Forecasting | *LightGBM, XGBoost, CatBoost, Scikit-Learn, Python* **2026**
- Engineered 40+ features including rolling statistics (7–360d), weekly lags, and price dynamics across 30,490 daily product-store hierarchical time series.
- Benchmarked 7 forecasting architectures via WRMSSE; LightGBM outperformed XGBoost, CatBoost, and LSTM baselines with a top score of 0.5666.
- Optimized training pipelines utilizing int8 memory quantization and GOSS sampling to efficiently process zero-inflated demand distributions.